

Geometric Scale Estimation using Multisensory Cues in Virtual Reality

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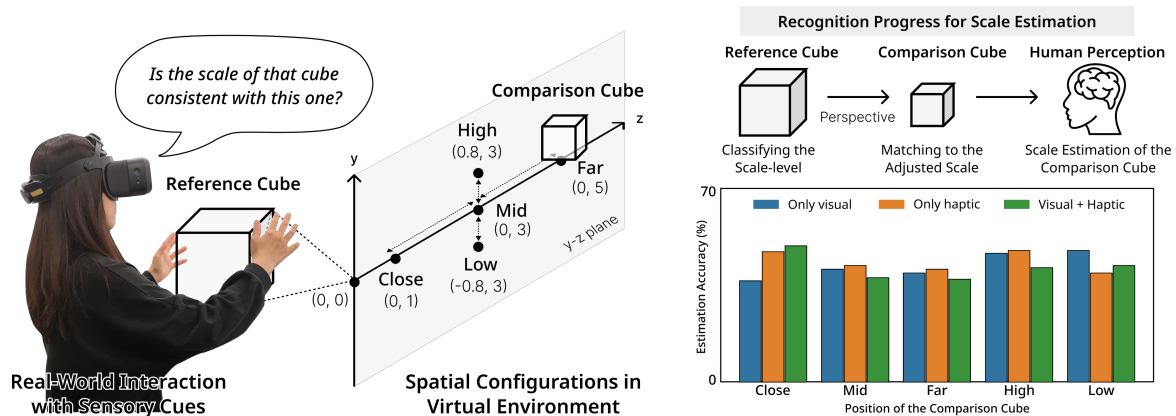


Figure 1: In our virtual environment, participants used sensory cues to interact with a reference cube and estimate the scale of a comparison cube placed at one of five positions varying in depth and height. They formed a mental model of the classified scale of reference cube and used perspective to assess whether it matched the comparison cube. Estimation was most accurate at closer distances, highlighting the effectiveness of multisensory cues in near space.

ABSTRACT

We present a study that investigates how multisensory integration influences three-dimensional object scale estimation in virtual reality (VR). The accurate scale perception remains a critical yet underexplored component of spatial cognition in VR, especially regarding vertical positioning, compared to the predominantly focused on horizontal depth. Although multisensory feedback has demonstrated clear advantages in real-world settings, the mechanism through which visual and haptic cues interact to support scale estimation in VR is not fully understood. In this study, we examine perceptual accuracy using 3-level scales of reference and comparison cubes placed at systematically varied horizontal depths and vertical heights, across three stimulus conditions such as visual-only, haptic-only, and combined visual-haptic feedback. We analyzed the main and interaction effects of those factors on estimation accuracy, as well as tendencies toward over- and underestimation. Results showed that spatial positions and sensory conditions had a greater influence on scale perception than the object's actual scale, with inconsistent estimates at vertical positions and improved accuracy under multisensory cues in close-range conditions. Our findings contribute to the theoretical understanding of cross-modal perception and offer practical implications for designing perceptually robust VR interfaces.

Index Terms: Multisensory cues, scale estimation, human perception, virtual reality.

1 INTRODUCTION

Virtual Reality (VR) has become an essential platform enabling immersive human-computer interaction, significantly impacting do-

mains like education, healthcare, design, and entertainment. As VR technology advances, research increasingly emphasizes understanding the perceptual processes critical for effective user experiences [1, 2]. A central yet insufficiently explored perceptual challenge in VR involves accurately judging object scale, which involves perceiving size, distance, and spatial relationships [3, 4].

Spatial perception integrates multiple sensory modalities, notably vision and touch, to form accurate spatial judgments [5, 6]. While multisensory integration is recognized for enhancing spatial accuracy in VR [7, 8], the precise mechanisms through which visual and haptic cues interact, particularly in spatial perception tasks, remain unclear [9, 10]. Existing research on spatial perception in VR predominantly addresses horizontal depth estimation, overlooking critical perceptual dynamics introduced by vertical positioning [11, 12]. Real-world interactions frequently involve vertically distributed objects—from overhead to floor-level placements—that present unique perceptual and motor challenges not yet systematically addressed in VR research [13, 14, 15].

Specifically, the integration of haptic feedback in spatial perception tasks involving varied vertical positions has received limited attention [16, 17]. Overhead interactions require upward visual attention and modified arm biomechanics, potentially affecting perceptual accuracy differently than interactions at floor-level, which involve downward attention and distinct motor strategies [18, 19].

This study addresses this critical gap by examining how combined visual and haptic feedback influences three-dimensional object scale estimation in VR. Under three sensory conditions, visual-only, haptic-only, and integration of visual and haptic, we investigate perceptual accuracy on scale estimation across varying size variations of the geometric cube, and spatial positions of horizontal depths and vertical heights.

To investigate scale perception within a large-scale spatial configuration, we defined the reference cube with dimensions of $20\text{cm} \times 20\text{cm} \times 20\text{cm}$. Based on this reference cube, three scale levels of the comparison cube, such as small, medium, and large, were established to assess differences in the accuracy of the estimation across varying cube scales. These levels were determined through

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a pilot study, which accounted for the minimum haptic threshold of the participants to detect scale differences, thus ensuring that each scale level exceeded the minimal visual discrimination threshold. Estimation fields were defined at five spatial positions with close, mid, far, high, and low. The high and low positions were designated based on angular visibility within a participant's initial field of view to reflect natural perceptual conditions in immersive VR.

In the main study, we examined how estimation accuracy varied depending on the stimulus condition, the reference cube's scale, and the spatial position of the comparison cube. Specifically, we analyzed which conditions showed the highest accuracy in scale estimation and investigated humans' tendencies toward underestimation or overestimation under different spatial and sensory configurations.

The result of scale estimation from participants was more strongly influenced by the spatial position of the comparison cube and the sensory condition than by the reference cube's scale itself. Multisensory feedback improved accuracy when the comparison cube was positioned at close range, while haptic-only conditions indicated higher accuracy at greater distances. Notably, in the multimodal condition, participants tended to underestimate the scale of distant objects, revealing a consistent perceptual bias under sensory conflict. These findings offer insight into which sensory configurations most effectively support accurate scale perception for distant objects in large-scale VR environments.

The outcomes of this research advance theoretical understanding of multisensory spatial perception and provide practical guidelines for designing perceptually accurate VR systems [20, 21], benefiting diverse fields including medical training, architectural visualization, industrial design, and rehabilitation therapies [22, 23].

In this paper, we present the following contributions:

- We investigated object scale perception by utilizing both visual and haptic cues, while also accounting for overestimation and underestimation tendencies.
- We employed a broader range of spatial configurations to more comprehensively cover variations in perceived scale, including vertical positions such as high and low.
- We proposed design guidelines for future VR environments that consider both sensory conditions and spatial positioning to support more accurate scale perception.

2 RELATED WORKS

2.1 Scale Perception in Virtual Reality

Scale perception research, which has been conducted in-depth based on the physical environment and 2D, has gradually been conducted in VR environments utilizing head-mounted displays (HMDs). Prominent among this wave of scholarly works were the investigations into the utility of virtual hands as a reference for object scale perception, explored by researchers such as Jung et al. [24] and Ogawa et al. [25]. Several results showed that people perceive the scale of objects by comparing them with their virtual hands in a VR environment, depending on the scale and realism of the virtual hands [26, 25]. In another study, subjects estimated the scale of objects in a VR environment of outer space, an alien landscape, and a typical cityscape, resulting in subjects estimating scale more accurately with familiar visual cues [27]. In addition, research shows that the scale of familiar objects in reality and virtual reality is different, and it highlights the importance of understanding the perception of scale in VR, which is different from reality [28]. In this way, most HMD-based research continued to prioritize visual perception [28, 29].

Although not only visual perception, object perception involves restricted attention, the induction of recognition, and practical

thinking [30]. It utilizes the perceptual abilities of other modalities, such as tactile and auditory senses. However, studies on scale perception in VR are still conducted mainly using visuals only, and there is a lack of studies that apply tactile perception. The present study identifies promising applications of VR environments and multisensory integration within scale perception research, suggesting theoretical understandings of perceptual processes.

2.2 Multisensory Cues in Object Recognition

Object recognition undergoes a complicated and prolonged adaptive process, reaching its full maturation through the integration of diverse sensory experiences [31]. Humans recognize objects and deduce their properties by integrating information from multiple sensory channels and incorporating feedback from motor commands [32]. In addition to unisensory findings, evidence supports the effective integration of visual and haptic inputs to enhance recognition performance [33, 34].

Visuo-haptic strategies demonstrate substantial performance improvements compared to each singular modality for object recognition [35]. Jung et al. found that multisensory feedback influenced participants' responses and confidence levels [36]. Another study reveals a pronounced search advantage when contextual learning occurs in a multisensory environment, indicating that multisensory stimulation improves visual search behavior [37]. These studies indicate that multisensory contributions are effective in improving object recognition. Although studies have incorporated haptic cues in object recognition, they have primarily focused on real-world situations [38, 39]. Additionally, limited research has investigated visual, haptic, and visual-haptic modalities individually in depth within VR environments.

2.3 Spatial Configuration for Scale Estimation

Previous studies have analyzed the perception of scale according to differences in distance [40, 41]. Although prior studies have focused on near-field interactions [42, 43, 44], the present study examines how multisensory environments affect human perception when objects are positioned far apart. Furthermore, the VR environment has not been explored for the scale perception of vertical direction. However, the VR environment has the potential to interact with objects in various positions, from near to far and from low to high, differently from the real world. Additionally, it is possible to interact with objects without directly touching them in VR [45]. Therefore, more in-depth research on human perception is required to provide an appropriate and precise VR environment in advance. Therefore, this research explores human perception under five conditions: horizontal depths (close, mid, far) and vertical heights (high, low).

3 DESIGN RATIONALE

Our research aims to investigate the influence of human visual and haptic perception on object scale judgment in a large-scale of environment in VR, particularly for varying scales of objects positioned at different vertical heights and horizontal depths. Before the experiment, we establish research questions, define virtual environmental settings, and outline the experimental procedures.

3.1 Research Questions on Scale Estimation

Based on the previous stimulus conditions, we set three research questions: 1) Does combined visual and haptic feedback produce significantly lower estimation errors compared to visual-only feedback for objects positioned at varying vertical heights and horizontal depths? 2) Is there a difference between accuracy in size estimation and distance estimation tasks across overhead, eye-level, and floor-level objects? and 3) Does multisensory integration compare to unimodal conditions systematically over any reference cube's scale variability? According to the above questions, we designated

the variations of the cube's scale and the spatial environment for our study.

3.2 Experimental Design

3.2.1 Geometric Object and Sensory Conditions

In this research, we defined a **Reference Cube** and a **Comparison Cube** to determine whether the comparison cube is smaller than, identical to, or larger than the reference cube in scale. The reference cube is always placed in front of the user and provides sensory cues, and the comparison cube appears in different positions in each trial. The reason why two virtual objects were selected as cubes was to eliminate potential biases related to the scale of real-world objects. When we perceive an object, we use our knowledge of its physical scale, which affects our judgment of its scale [46]. Scale information is only used for perception when the actual scale of the object is precise, and the previous experiment studied how subjects perceive scale information using geometric shapes with no apparent physical scale, such as cubes [38].

And the sensory conditions for the cube's scale estimation used in our study are as follows. For the scale estimation progress in the main study, two cubes per stimulus are presented sequentially, one as a reference cube and the other as a comparison cube (see Fig. 2). The reference cube is consistently positioned at a fixed location with one of three levels of scales (small, medium, large). The comparison cube is displayed at one of five locations, with one of three sizes (smaller than, identical to, or larger than the reference object).

- Only visual (V): Participants view the reference and comparison cubes without haptic feedback.
- Only haptic (H): Participants judge cube size by touching only the reference cube with visual input of the comparison cube.
- Visual + Haptic (VH): Participants experience haptic feedback of the reference cube and visual inputs of the reference and comparison cubes.

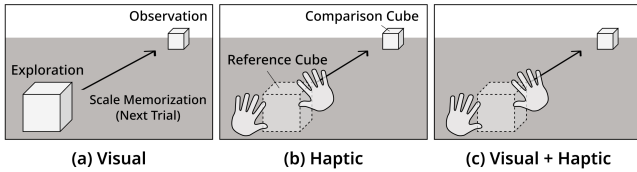


Figure 2: The three stimulus conditions used in this study. A human first explores the reference cube and memorizes its scale, then observes the comparison cube to estimate its scale in the next trial. In each condition, the reference cube is explored using (a) only visual, (b) haptic feedback with invisible hands and cube, and (c) both visual and haptic feedback with invisible hands and cube.

3.2.2 Virtual Environment Settings

Cube Scale Variations In prior work, the far distance in actual physical space experiments was unreachable with 120cm [40]. In the current VR experimental environment, the distance variation we specified was more expansive, so we increased the scale of the reference object with 20cm $\times$ 20cm $\times$ 20cm as a medium scale.

To divide the reference cube into three scale levels, it is necessary to determine the minimum scale differences at which users can visually and haptically distinguish between the small and large scales relative to the medium scale. However, there was a lack of research on the threshold that users distinguish when using only haptics. Since it is evident that if the minimum scales of small

and large cubes are distinguishable by humans based on the scale of the reference cube, experiencing smaller and larger cube scales through haptics is naturally guaranteed. Therefore, in this study, we conducted preliminary research to investigate the scale threshold that humans distinguish when using only haptics by comparing the small and large scales of cubes with the medium scale cube. The experiment's results were used to determine the scale of the reference cubes in the main study.

Estimation Fields We designed the virtual environment within a specific large range of levels to enable the perception of three-dimensional virtual objects with various scales through these sensory modalities. Horizontal depths of the estimation fields were divided into three levels; *Close* (1m), *Mid* (3m), and *Far* (5m) to analyze the user's scale estimation results based on the difference in distance between objects, as can be seen in Fig. 3. The *High* and *Low* conditions were applied at mid-distance to place the comparison object in the initial field of view without tilting the head.

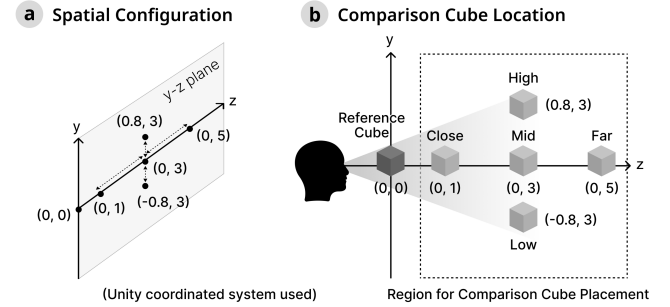


Figure 3: Spatial configuration used in the VR environment. (a) Five spatial positions are defined based on the origin, with all positions aligned at $x = 0$. (b) The possible regions where the comparison can be located, and the reference cube is always placed in front of the observer.

Vertical height used in this study has three levels as *High* (+0.81m), *Mid* (0m), and *Low* (-0.81m). The height standard is the human's eye-level, and the reason for the three height classifications is based on the useful field of view (UFOV) that can be recognized while the eyes and head are fixed without movement [47]. Prior studies have shown that the UFOV of young adults can effectively process information up to about 30 degrees [47, 48]. Based on these results, this experiment was designed to present stimuli within a range of 3m in front of the subject and 0.81m above and below the subject so that all visual stimuli within the UFOV of young adults could be processed.

Therefore, the scenario in which object scale estimation is conducted using sensory cues within the estimation fields of the VR. Since our goal is to analyze object scale estimation in large-scale virtual environments, only the reference cube provides sensory cues, while the comparison cube is available for visual observation only.

4 EXPERIMENTS

All participants wore a Varjo VR-3 on their heads and HaptX gloves on both hands for the pilot and main user study. We recruited 35 unique participants, including 10 participants (5 males, mean age of 26.00, $\sigma = 2.36$) for the pilot study, 25 participants (10 males, mean age of 26.64, $\sigma = 2.89$) for the main study. Fig. 4 presents the equipment configuration employed during the study.

4.1 Pilot Study: Employing the Discrimination Threshold on Cube's Scale with Haptic Feedback

To ensure the perception of changes in the designated cube's scale with haptic feedback, we figured out which threshold is adequate

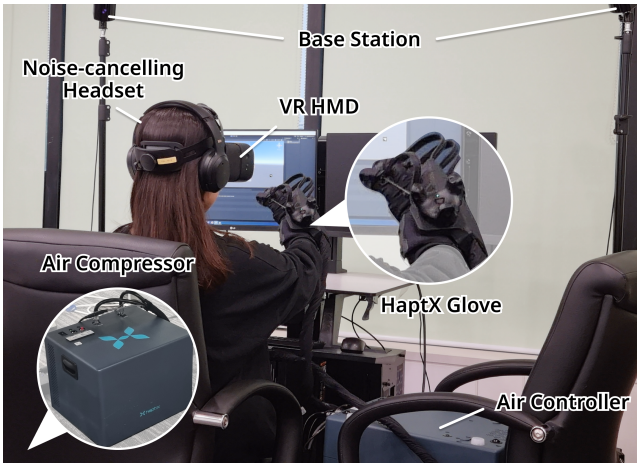


Figure 4: Hardware setup for the user study. The air compressor and controller are configured to support pneumatic-based haptic gloves. (DK2, HaptX Inc.)

for discriminating scales before conducting the main cube-scale estimating study. We explored the perceptual distinction on an increased or decreased scale of the cube using a Just Noticeable Difference (JND) method.

A prior study primarily used three scale levels to determine the scale of an object: 50 % of the volume of the reference object (100 %) as a small scale and 150 % of the reference one as a large scale [38]. In another experiment, they fixed the diameter of the reference object (sphere) at 50mm and randomized the scale of the comparison object between 38mm and 62mm in increments of 1mm [40]. Accordingly, in this research, we determined the scale of the increasing or decreasing cube based on the length of one side of the reference cube. The starting cube scales were defined as having each side length equal to 50% and 150% of the reference cube. The initial step size was set to 2cm and was halved each time a participant gave an incorrect response.

4.1.1 Experimental Procedures

In the VR environment, all participants experienced two conditions: one in which the cube started at a larger scale ($30\text{cm} \times 30\text{cm} \times 30\text{cm}$) and gradually decreased in size, and another in which it started at a smaller scale ($10\text{cm} \times 10\text{cm} \times 10\text{cm}$) and gradually increased. For the decreasing condition, a two-down one-up staircase method was used, whereas for the increasing condition, a two-up one-down staircase was applied.

Within each trial, participants experienced sequentially a reference cube and a transforming cube with a side altered gradually. During this procedure, both cubes and participants' hands were invisible, satisfying the participants relied solely on the haptic feedback received from the two cubes (can be seen in Fig. 5). The reference cube, which served as the constant baseline for all comparisons, consistently appeared $20\text{cm} \times 20\text{cm} \times 20\text{cm}$. Participants were given enough exposure time of 6 seconds each to experience both the reference cube and the subsequent increasing or decreasing cube. Trials were continued until four reversal points were collected, which is considered a sufficient number for reliable threshold estimation.

4.1.2 Pilot Results & Key Findings

Table 1 summarizes the mean JND values and the average number of trials recorded for each condition. The results indicate a higher perceptual threshold for detecting a decrease from a larger cube compared to detecting an increase from a smaller cube.

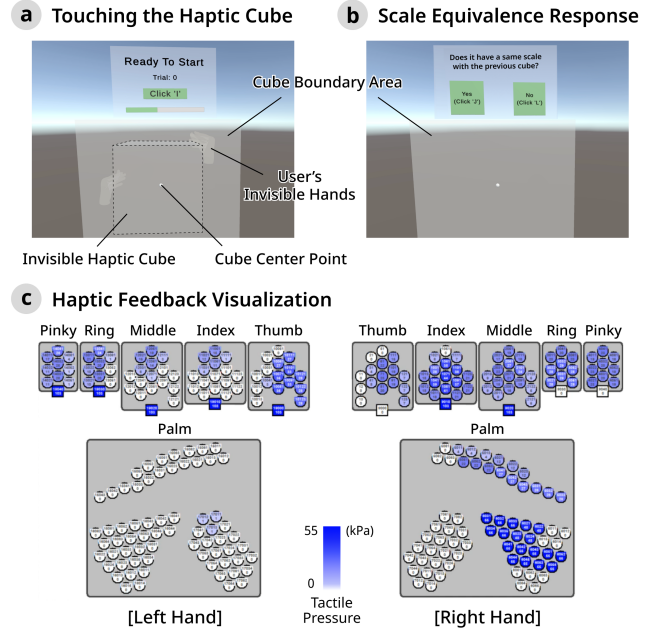


Figure 5: This illustrates the procedure of the pilot study. (a) Participants estimate the scale by touching the cube within the gray boundary. (b) They verbally respond with 'yes' or 'no' to indicate whether the scale matches the previous cube. (c) Automated visualization of the haptic feedback provided by the glove while touching the reference cube.

Condition	Mean JND (SD) (cm)	Mean Trials (SD)
Decreasing from a larger cube	26.18 (2.41)	20.40 (3.69)
Increasing from a smaller cube	17.43 (1.06)	22.80 (3.33)

Table 1: Mean JND values representing one side length of a distinguishable cube and average number of trials per condition

Based on the results obtained from the pilot study, we observed that the condition that is increasing from a smaller cube required more trials and resulted in a cube scale closer to that of the reference cube, compared to decreasing from a larger cube. Thus, we found that participants are more sensitive and easily perceive changes when the cube scale increases rather than when it decreases.

We also determined the dimensions of the reference cubes to be used in the main study as follows: the small-scale cube has a side length of 17.43cm, the medium-scale cube 20cm, and the large-scale cube 26.18cm as shown in Fig. 6.

4.2 Main Study: Scale Detection of the Cube in VR

To estimate the cube's scale with multisensory cues, we conducted a total of 135 trials per participant. It contains 3 sensory cues (V, H, VH) $\times$ 5 positions where the comparison cube would be placed (as seen in Fig. 3 (b)) $\times$ 3 scale variations of the reference cube (small, medium, large) $\times$ 3 scale variations of the comparison cube (smaller, identical, larger). Using a within-subjects design, we proceeded with the study based on a balanced Latin square method.

4.2.1 Main Experiment Flow

Within the VR scene designed for the study, participants sequentially experience the reference object and comparison object according to the given sensory condition [40, 49, 50, 51] as seen in Fig. 7. The reference cube was positioned slightly below the par-

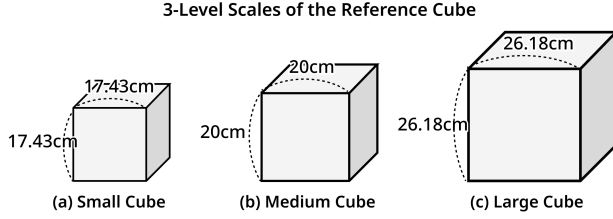


Figure 6: Three scale variations defined for the reference cube, which have small, medium, and large scales.

participant’s eye level to facilitate hand interaction, and this setup was calibrated individually for each participant prior to the start of the study. Participants press the ‘smaller’ button if they perceive the comparison cube to be smaller than the reference cube, the ‘same’ button if they perceive as equal in scale, and the ‘larger’ button if they perceive as larger. There is no time limit for responses, and each trial concludes immediately upon their input. However, we instructed them to respond as quickly and accurately as possible.

The reason why we placed three buttons was to gain deeper insight into participants’ scale estimation errors under various conditions, allowing us to determine not only whether their estimations were incorrect but also whether they underestimated or overestimated the cube’s scale. Also, since an environment with vanishing points is required to recognize the scale and position of objects, Unity’s default environment with a horizon line was used.

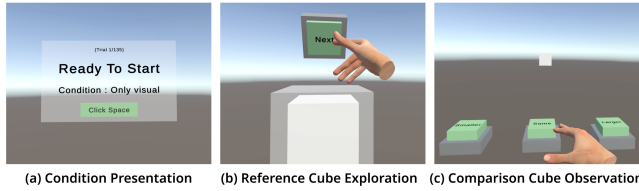


Figure 7: This shows the procedure of the main study. (a) A stimulus condition is presented. (b) Depending on the condition, participants experience the reference cube either visually or haptically. (c) While visually observing the comparison cube placed at a distance, they indicate its perceived scale relative to the reference cube by selecting one of the buttons.

In the ‘Only haptic’ condition, participants were informed in advance that the first point at which haptic feedback was perceived when bringing their hands together outside the gray boundary indicated the border of the reference cube. They were instructed to estimate the cube’s scale based on this border. It was also noted that stronger penetration into the cube’s border would result in higher reaction forces, naturally producing stronger haptic feedback.

4.2.2 Evaluation and Analysis

We set the baseline group for evaluation as a condition with ‘Only visual’, *Medium* scale of the reference cube, and spatial position *Close* of the comparison cube.

Fig. 8 visualizes the accuracy distributions for each stimulus condition, separately depicting the effects of the two independent variables. In Fig. 8 (a), under the baseline condition using ‘Only visual’ cue with a *Medium* reference cube, accuracy decreases when the cube is either smaller or larger in scale. In the ‘Only haptic’ condition, accuracy increases for both small and large scales compared to the *Medium* scale. For the multisensory condition combining both visual and haptic cues, accuracy tends to decrease as the scale of the cube increases. Meanwhile, in Fig. 8 (b), for comparison cube positioned near the participant and approximately at eye level,

which is *Close*, accuracy followed the order of ‘Only visual’, ‘Only haptic’, and ‘Visual + Haptic’. As the comparison cube moved farther away, such as positions *Mid* and *Far*, accuracy shifted, with ‘Only haptic’ outperforming ‘Only visual’, and ‘Visual + Haptic’ showing the lower performance. At positions outside the typical eye-level height, such as *High* and *Low*, the use of sensory cues appeared less consistent compared to other positions.

To examine the effects of sensory condition, scale variation in the reference cube, and spatial position of the comparison cube on response accuracy, we conducted a binary logistic regression based on whether the responses were correct or not as a dependent variable. All independent variables were also categorical predictors. The logistic regression model did not show overall statistical significance for three independent variables ($p > .05$). However, when the interaction terms were added, several significant effects emerged in the post-hoc analysis.

Variable	β	SE	z	p	OR
Intercept	-0.575	0.241	-2.392	.017	0.56
VH	0.762	0.334	2.282	.022	2.14
Low	0.816	0.335	2.440	.015	2.26
H $\times$ Mid	-0.947	0.475	-1.995	.046	0.38
VH $\times$ Mid	-1.275	0.477	-2.676	.007	0.27
VH $\times$ Far	-1.046	0.475	-2.203	.028	0.35
H $\times$ Low	-1.086	0.469	-2.315	.021	0.33
VH $\times$ Low	-1.030	0.468	-2.201	.028	0.35

Table 2: Logistic regression results for selected predictors based on the full model including main effects and interactions. β represents the estimated regression coefficient (log-odds), SE is the standard error of the coefficient, and OR is the odds ratio obtained by exponentiating the coefficient ($\exp(\beta)$).

In Tab. 2, the intercept was significantly negative ($\beta = -0.575, p = .017$), indicating that the baseline condition was associated with performance below chance level (odds ratio ≈ 0.56 with 36% accuracy). A significant main effect of the stimuli condition was indicated for ‘Visual + Haptic’ and comparison cube’s position was observed for *Low*, suggesting that this condition or location facilitated more accurate responses relative to the baseline. Critically, an interaction between the ‘Visual + Haptic’ condition revealed an accuracy drop with position *Mid*, *Far*, and *Low*. Similarly, the ‘Only haptic’ condition shows a significant effect ($p < .05$) with a position of *Mid* or *Low* by decreasing the accuracy. The variation in the scale of the reference cube does not appear to significantly affect participants’ performance across different conditions. These findings indicate that the impact of stimulus condition on accuracy was modulated by the spatial position, compared to the baseline.

As shown in Fig. 9, regardless of the stimulus condition, we observed a general trend where overestimation decreases and underestimation increases as the comparison cube moves from the *Close* to the *Mid* and *Far* position. Additionally, when the comparison cube is located in the *Close* position, the presence of haptic cues appears to contribute more expressively to correct responses.

To examine deeply how perceptual judgment types (correct, overestimation, underestimation) were influenced by stimulus condition, scale of the reference cube, and position of the comparison cube, we conducted a multinomial logistic regression. The dependent variable was the participant’s judgment type, and the baseline was set with ‘Correct’ answer, a medium scale of the reference cube, and a close position of the comparison cube. For the main effects, several spatial configurations and *Large* scale of the reference cube significantly influenced the likelihood of producing overestimations or underestimations. Specifically, the probability of producing an overestimation was significantly lower at *Far* ($\beta = -0.445, p = .001$), *High* ($\beta = -0.410, p = .002$), and

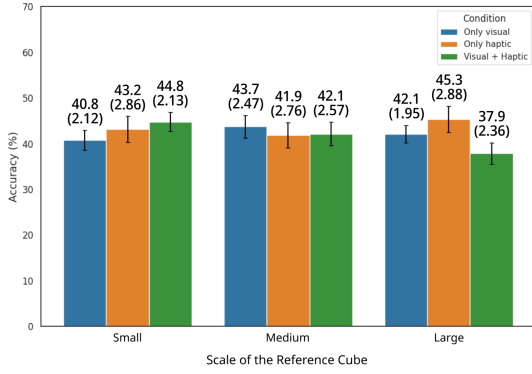
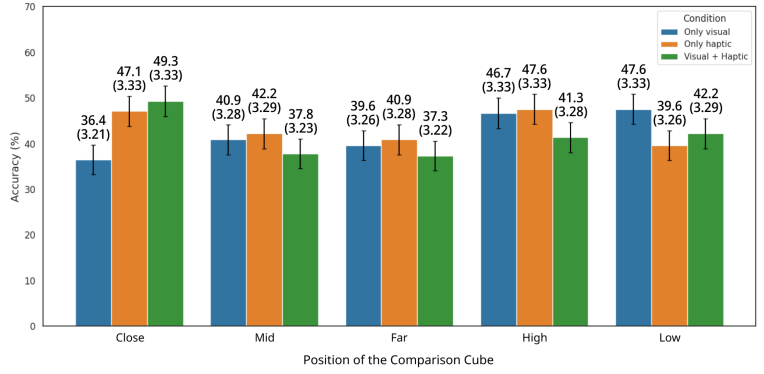
a Accuracy by Scale of the Reference Cube**b Accuracy by Spatial Position of the Comparison Cube**

Figure 8: This presents accuracy graphs according to the stimulus conditions. The value above each graph shows the mean, with the standard error shown in parentheses. (a) The accuracy of correct responses is determined by the scale of the reference cube. (b) The accuracy of correct responses is determined by the spatial position of the comparison cube.

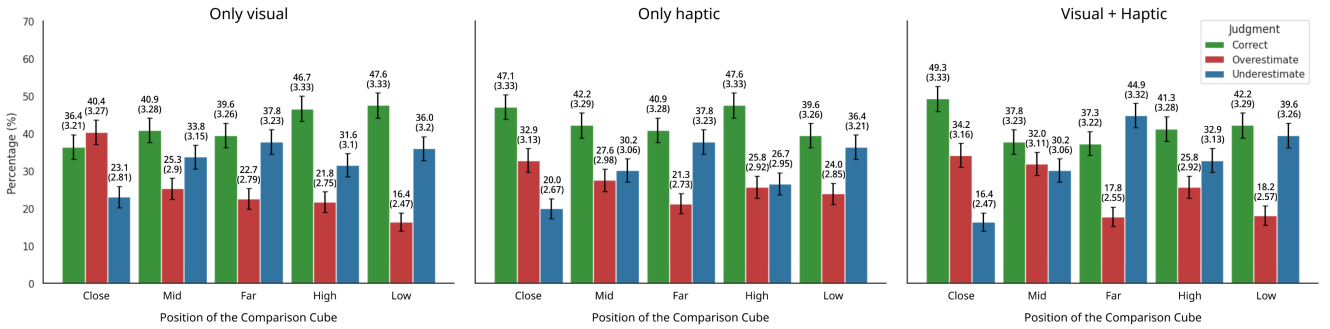
Detailed Scale Judgment of the Comparison Cube

Figure 9: This illustrates the proportion of correct, overestimation, and underestimation responses for each stimulus condition, depending on the spatial position of the comparison cube. The value above each graph shows the mean, with the standard error shown in parentheses.

Low positions ($\beta = -0.590, p < .001$) of the comparison cube, and when the reference cube's scale was *Large* ($\beta = -0.375, p = .001$). Conversely, the probability of underestimating was significantly higher at *Mid* ($\beta = 0.560, p < .001$), *Far* ($\beta = 0.837, p < .001$), *High* ($\beta = 0.413, p = .003$), and *Low* ($\beta = 0.670, p < .001$), and with the *Large* reference cube ($\beta = 0.318, p = .001$). This finding aligns with the trends in overestimation and underestimation observed in Fig. 9, further supporting the observation that distance has a more pronounced effect than scale variation. Sensory condition ('Only haptic' or 'Visual + Haptic') did not have a statistically significant effect compared to 'Only visual'.

In the post-hoc analysis, a multinomial logistic regression revealed significant interactions (shown as Tab. 3). Participants were less likely to overestimate the *Large* reference cube compared to the *Medium* scale. The likelihood of overestimation significantly decreases at all positions beyond the baseline *Close* position (0, 1). Significant interaction effects were also observed between 'Visual + Haptic' and certain positions of the comparison cube. These results suggest that in the 'Visual + Haptic' condition, the possibility of overestimation increased again at those positions (*Mid*, *High*, and *Low*). In terms of underestimation, participants were more likely to underestimate when the reference cube's scale is *Large* compared to the *Medium* scale. Interaction effects again showed that underestimation was amplified in the 'Visual + Haptic' condition at specific positions *Far*, *High*, and *Low*.

4.2.3 Post-Experiment Interview

After completing the study, we asked participants the following questions for the in-depth interview. We requested their estimation progress by utilizing sensory cues in terms of contrast between a reference cube and a comparison cube. Participants provided a range of qualitative responses regarding their perceptual strategies when interacting with the reference cube and estimating the comparison cube using visual, haptic, and combined modalities.

First, participants utilized both visual and haptic cues actively to classify the scale of the reference cube according to the stimulus condition, forming an internal scale that was later used to estimate the scale of the comparison cube (P2, P6-7, P14-16, P18-P25). They often categorized the reference cube into discrete levels (e.g., small, medium, large) and mentally projected this to the comparison cube using perspective and a vanishing point. The number of levels varied across participants, ranging from two to four, with visual categorizations generally being more precise.

Second, visual cues were reported as more dominant overall, especially when the comparison cube appeared farther away (P3-4, P8, P11, P15, P17, P23). Since the comparison cube was naturally observed through vision, participants sometimes compensated for perceived distortions caused by depth or spatial layout. While haptic cues were secondary, they enhanced confidence and helped validate visual impressions when both modalities were available (P2-3, P6-8, P11-15, P17, P19-20, P22-24). However, several participants noted mismatches between modalities, where visual and haptic per-

Variable	β	SE	z	p	OR
Large	-0.506	0.197	-2.573	.010	0.60
Mid	-0.598	0.229	-2.619	.009	0.54
Far	-0.680	0.234	-2.909	.004	0.50
High	-0.883	0.232	-3.809	.000	0.41
Low	-1.188	0.246	-4.842	.000	0.30
H $\times$ Low	1.042	0.337	3.095	.002	2.83
VH $\times$ Mid	0.795	0.316	2.515	.012	2.21
VH $\times$ High	0.772	0.322	2.398	.016	2.16
VH $\times$ Low	0.708	0.342	2.068	.039	2.02
Large	0.425	0.169	2.516	.012	1.53
VH $\times$ Far	0.885	0.337	2.626	.009	2.42
VH $\times$ High	0.808	0.341	2.368	.018	2.24
VH $\times$ Low	0.861	0.335	2.572	.010	2.37

Table 3: Multinomial logistic regression results for selected predictors based on the full model including main effects and interactions. The top section is related to overestimation, and the bottom is about underestimation.

ceptions of the same cube differed, sometimes causing confusion or longer decision times (P2, P5, P9, P25). Some also reported that the ‘Only haptic’ condition led them to unintentionally underestimate the cube’s scale, in contrast to when visual cues were provided (P4, P25).

Third, in terms of the position or scale of the cubes, participants reported feeling more confident in estimating the comparison cube’s scale when it was positioned closer. On the contrary, they relied more on intuition based on sensory cues when the comparison cube was farther away (P2, P4, P11, P13, P16-17, P24), and this tendency to rely on intuitive estimation was more pronounced when the cube was located at *High* or *Low* positions (P1, P7, P11). And since participants primarily relied on perspective and depth cues to judge scale, they noted that larger cubes were easier to differentiate due to more noticeable scale contrasts (P16, P23).

5 DISCUSSION

The present findings provide insights into the influence of multisensory cues, spatial configuration, and object scale on human scale perception. Although overall logistic regression models showed weak main effects, notable interactions, and multinomial analyses revealed adaptive perceptual strategies under complex sensory-spatial conditions.

5.1 Interpretation of Key Findings

Performance of Spatial Layout Participants performed below chance level under the baseline condition, which has ‘Only visual’ condition, *Medium* scale of the reference cube, and *Close* position of the comparison cube. Specifically, spatial positions such as *Mid*, *Far*, and *Low* exhibited reduced accuracy, suggesting increased difficulty in scale estimation as the comparison cube was positioned further away or vertically aligned. These spatial influences were supported by the multinomial models, where overestimation was less likely and underestimation more likely at *Far*, *High*, and *Low* positions. These results highlight the importance of egocentric spatial relations, particularly distance and vertical positioning, in shaping scale judgments.

Interaction Between Sensory Modalities and Spatial Position Integration of visual and haptic cues improved scale estimation performance at close distances. In near-space contexts, haptic feedback may have reinforced participants’ perceptual confidence in object scale, facilitating more precise judgments. However, in distant or vertically challenging positions such as *Far*, *High*, and

Low, combining visual and haptic cues appeared to mismatch, possibly introducing cognitive overload or perceptual conflicts. Under these conditions, reliance solely on haptic input or providing enough time for visual and haptic cues to perceptually integrate could be beneficial for more consistent and stable perceptual judgments.

Influence of Object Scale on Perceptual Bias Participants tended to underestimate *Large* reference cubes, reflecting a cognitive normalization strategy to anchor perceptions within a familiar scale range. This underestimation bias intensified in the ‘Visual + Haptic’ condition, especially under difficult spatial placements such as *Far*, *High*, and *Low*. This phenomenon highlights complex interactions between object geometry and sensory integration.

Multinomial analysis further revealed that overestimation was least likely in spatially extreme conditions, reflecting a conservative estimation strategy when depth or vertical cues were limited. Conversely, underestimation was prevalent with large-scale references and non-central cube placements, possibly due to increased perceptual compression or difficulty in mentally projecting size along depth and height axes. Interestingly, these biases were more strongly modulated by spatial configuration than by object scale, implying that participants relied more on spatial heuristics when interpreting ambiguous or complex spatial layouts.

5.2 Design Guidelines for Scale Estimation in VR

Based on the observed trends from our research, several design guidelines can be derived for applications requiring accurate scale estimation in immersive environments.

Support Close-range Interactions for Critical Judgments Scale estimation is significantly more accurate at closer distances using multisensory cues. When precise size judgments are essential, such as training simulations and virtual prototyping, key objects should be placed within near reach of the user to leverage stronger spatial and sensory integration.

Place Surrounding Objects in Spatially Extreme Positions In cases such as *Mid*, *Far*, *High*, and *Low* positions, providing surrounding objects nearby for relative size comparison may be useful. Since large objects tend to be underestimated at these spatial positions, placing similarly scaled reference objects or familiar standardized elements (e.g., building doors or windows) nearby could help prevent the underestimation or overestimation problems.

Strategic Use of Sensory Cues in Complex Layouts In conditions with large spatial separation or vertical displacement, relying solely on haptic input may reduce conflict and cognitive load. It can be more reliable for evaluating scale in large-scale VR environments such as architectural walkthroughs, industrial manufacturing, and short-range teleoperation tasks. However, when multisensory cues are available, sufficient time might be required to enhance anchoring mechanisms that improve visual-haptic integration with compensating for perceptual degradation.

5.3 Limitations and Future Work

Even though it offers valuable insights into perceptual biases and sensory integration, this research has several limitations that highlight areas for future investigation.

The spatial configurations explored here were restricted to discrete positions, omitting continuous movement or dynamic interactions typically encountered in real-world AR/VR environments. Thus, future studies could incorporate more dynamic and realistic spatial interactions. The vertical positions (*High* and *Low*) tested were also limited and can constrain generalization about spatial effects on object-scale perception, so future work can broaden this spatial range.

Additionally, complexities involving various object shapes, textures, and lighting conditions were not considered. We encourage

investigating these factors that have potential nuanced effects on scale perception represents an important direction for future research. And haptic feedback was delivered using a single type of device. We think subsequent studies can explore various haptic mechanisms to test the robustness and generalizability of these findings.

6 CONCLUSION

This study explored geometric scale estimation in VR by manipulating stimulus conditions, the scale of a reference cube, and the spatial position of the comparison cube. Since we focused on large-scale perception, we designed a 3-level depth configuration which was extended up to 5m and indicated vertical variation by positioning the cubes at high and low levels, offset by 0.81m from eye-level.

From the user study, when experiencing the reference cube according to the stimulus condition first, participants primarily relied on sensory feedback to classify the reference cube's scale mentally. Then, when estimating the scale of a distant comparison cube, they often projected the estimated scale of the reference cube to the target position for scale matching with a comparison cube. In the close distance, multisensory cues (visual and haptic) showed the highest accuracy, suggesting that haptic cues enhanced both estimation confidence and correctness. However, at far distances, modality mismatches between visual and haptic information degraded performance. The use of haptic feedback alone enhanced the accuracy of scale estimation, suggesting a growing potential for haptic reliance and the future utility of haptic modalities. At high and low vertical positions, participants showed increased uncertainty and relied more on intuition in estimations. Additionally, large-scale cubes were frequently underestimated, likely due to exaggerated perspective distortion when applying vanishing point projection in mental scaling.

These findings suggest that scale estimation is influenced more by spatial position and sensory modality than by object scale itself. The insights from this research offer practical implications for designing VR interfaces and applications, particularly within large-scale virtual environments.

REFERENCES

- [1] I. E. Sutherland *et al.*, "The ultimate display," in *Proceedings of the IFIP Congress*, vol. 2, no. 506-508. New York, 1965, pp. 506-508. 1
- [2] M. Slater and M. V. Sanchez-Vives, "Enhancing our lives with immersive virtual reality," *Frontiers in Robotics and AI*, vol. 3, p. 74, 2016. 1
- [3] R. S. Renner, B. M. Velichkovsky, and J. R. Helmert, "The perception of egocentric distances in virtual environments-a review," *ACM Computing Surveys (CSUR)*, vol. 46, no. 2, pp. 1-40, 2013. 1
- [4] J. W. Kelly, "Distance perception in virtual reality: A meta-analysis of the effect of head-mounted display characteristics," *IEEE transactions on visualization and computer graphics*, vol. 29, no. 12, pp. 4978-4989, 2022. 1
- [5] I. Rock and J. Victor, "Vision and touch: An experimentally created conflict between the two senses," *Science*, vol. 143, no. 3606, pp. 594-596, 1964. 1
- [6] M. O. Ernst and M. S. Banks, "Humans integrate visual and haptic information in a statistically optimal fashion," *Nature*, vol. 415, no. 6870, pp. 429-433, 2002. 1
- [7] S. Gepshtein and M. S. Banks, "Viewing geometry determines how vision and haptics combine in size perception," *Current Biology*, vol. 13, no. 6, pp. 483-488, 2003. 1
- [8] S. Lacey and K. Sathian, "Visuo-haptic multisensory object recognition, categorization, and representation," *Frontiers in psychology*, vol. 5, p. 730, 2014. 1
- [9] G. Desmarais, M. Meade, T. Wells, and M. Nadeau, "Visuo-haptic integration in object identification using novel objects," *Attention, Perception, & Psychophysics*, vol. 79, pp. 2478-2498, 2017. 1
- [10] M. Gonzalez-Franco and J. Lanier, "Model of illusions and virtual reality," *Frontiers in psychology*, vol. 8, p. 1125, 2017. 1
- [11] T. L. Yang, M. W. Dixon, and D. R. Proffitt, "Seeing big things: Overestimation of heights is greater for real objects than for objects in pictures," *Perception*, vol. 28, no. 4, pp. 445-467, 1999. 1
- [12] M. W. Dixon and D. R. Proffitt, "Overestimation of heights in virtual reality is influenced more by perceived distal size than by the 2-d versus 3-d dimensionality of the display," *Perception*, vol. 31, no. 1, pp. 103-112, 2002. 1
- [13] V. Interrante, B. Ries, and L. Anderson, "Distance perception in immersive virtual environments, revisited," in *IEEE virtual reality conference (VR 2006)*. IEEE, 2006, pp. 3-10. 1
- [14] B. R. Kunz, S. H. Creem-Regehr, and W. B. Thompson, "Testing the mechanisms underlying improved distance judgments in virtual environments," 2015. 1
- [15] J. K. Stefanucci and D. R. Proffitt, "The roles of altitude and fear in the perception of height," *Journal of Experimental Psychology: Human Perception and Performance*, vol. 35, no. 2, p. 424, 2009. 1
- [16] M. Girondini, M. Montanaro, and A. Gallace, "Exploring virtual reality object perception following sensory-motor interactions with different visuo-haptic collider properties," *Scientific Reports*, vol. 14, no. 1, p. 10011, 2024. 1
- [17] G. Buckingham, "Examining the size-weight illusion with visuo-haptic conflict in immersive virtual reality," *Quarterly Journal of Experimental Psychology*, vol. 72, no. 9, pp. 2168-2175, 2019. 1
- [18] Y. Ban, T. Narumi, T. Tanikawa, and M. Hirose, "Modifying perceived size of a handled object through hand image deformation," *Presence: Teleoperators and Virtual Environments*, vol. 22, no. 3, pp. 255-270, 2013. 1
- [19] J. Bergström, A. Mottelson, and J. Knibbe, "Resized grasping in vr: Estimating thresholds for object discrimination," in *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*, 2019, pp. 1175-1183. 1
- [20] S. Triberti, C. Sapone, and G. Riva, "Being there but where? sense of presence theory for virtual reality applications," *Humanities and Social Sciences Communications*, vol. 12, no. 1, pp. 1-6, 2025. 2
- [21] D. Freeman, S. Reeve, A. Robinson, A. Ehlers, D. Clark, B. Spanlang, and M. Slater, "Virtual reality in the assessment, understanding, and treatment of mental health disorders," *Psychological medicine*, vol. 47, no. 14, pp. 2393-2400, 2017. 2
- [22] C. Botella, J. Fernández-Álvarez, V. Guillén, A. García-Palacios, and R. Baños, "Recent progress in virtual reality exposure therapy for phobias: a systematic review," *Current psychiatry reports*, vol. 19, pp. 1-13, 2017. 2
- [23] S. G. Neri, J. R. Cardoso, L. Cruz, R. M. Lima, R. J. De Oliveira, M. D. Iversen, and R. L. Carregaro, "Do virtual reality games improve mobility skills and balance measurements in community-dwelling older adults? systematic review and meta-analysis," *Clinical Rehabilitation*, vol. 31, no. 10, pp. 1292-1304, 2017. 2
- [24] S. Jung, G. Bruder, P. J. Wisniewski, C. Sandor, and C. E. Hughes, "Over my hand: Using a personalized hand in vr to improve object size estimation, body ownership, and presence," in *Proceedings of the Symposium on Spatial User Interaction*. Berlin Germany: ACM, Oct. 2018, p. 60-68. [Online]. Available: <https://dl.acm.org/doi/10.1145/3267782.3267920> 2
- [25] N. Ogawa, T. Narumi, and M. Hirose, "Virtual hand realism affects object size perception in body-based scaling," in *2019 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. Osaka, Japan: IEEE, Mar. 2019, p. 519-528. [Online]. Available: <https://ieeexplore.ieee.org/document/8798040/> 2
- [26] S. A. Linkenauger, M. Leyrer, H. H. Bühlhoff, and B. J. Mohler, "Welcome to wonderland: The influence of the size and shape of a virtual hand on the perceived size and shape of virtual objects," *PloS one*, vol. 8, no. 7, p. e68594, 2013. 2
- [27] H. Park, N. Faghihi, M. Dixit, J. Vaid, and A. McNamara, "Judgments of object size and distance across different virtual reality environments: A preliminary study," *Applied Sciences*, vol. 11, no. 23, p. 11510, 2021. 2
- [28] A. M. Rzepka, K. J. Hussey, M. V. Maltz, K. Babin, L. M. Wilcox, and J. C. Culham, "Familiar size affects perception differently in virtual

- reality and the real world,” *Philosophical Transactions of the Royal Society B*, vol. 378, no. 1869, p. 20210464, 2023. 2
- [29] E. Blini, C. Desoche, R. Salemme, A. Kabil, F. Hadj-Bouziane, and A. Farnè, “Mind the depth: Visual perception of shapes is better in peripersonal space,” *Psychological science*, vol. 29, no. 11, pp. 1868–1877, 2018. 2
- [30] C. O’Callaghan, “Objects for multisensory perception,” *Philosophical Studies*, vol. 173, no. 5, pp. 1269–1289, 2016. 2
- [31] G. Purpura, G. Cioni, and F. Tinelli, “Development of visuo-haptic transfer for object recognition in typical preschool and school-aged children,” *Child Neuropsychology*, vol. 24, no. 5, pp. 657–670, 2018. 2
- [32] A. Sciutti, F. Damonte, M. Alloisio, and G. Sandini, “Visuo-haptic exploration for multimodal memory,” *Frontiers in Integrative Neuroscience*, vol. 13, p. 15, 2019. 2
- [33] S. Kim and T. W. James, “Enhanced effectiveness in visuo-haptic object-selective brain regions with increasing stimulus salience,” *Human brain mapping*, vol. 31, no. 5, pp. 678–693, 2010. 2
- [34] S. Kim, R. A. Stevenson, and T. W. James, “Visuo-haptic neuronal convergence demonstrated with an inversely effective pattern of bold activation,” *Journal of Cognitive Neuroscience*, vol. 24, no. 4, pp. 830–842, 2012. 2
- [35] T. Corradi, P. Hall, and P. Irvani, “Object recognition combining vision and touch,” *Robotics and biomimetics*, vol. 4, no. 1, pp. 1–10, 2017. 2
- [36] S. Jung, A. L. Wood, S. Hoermann, P. L. Abhayawardhana, and R. W. Lindeman, “The impact of multi-sensory stimuli on confidence levels for perceptual-cognitive tasks in vr,” in *2020 IEEE Conference on Virtual Reality and 3D User Interfaces (VR)*. IEEE, 2020, pp. 463–472. 2
- [37] S. Chen, Z. Shi, H. J. Müller, and T. Geyer, “Multisensory visuo-tactile context learning enhances the guidance of unisensory visual search,” *Scientific Reports*, vol. 11, no. 1, p. 9439, 2021. 2
- [38] D. E. Holler, M. Behrmann, and J. C. Snow, “Real-world size coding of solid objects, but not 2-d or 3-d images, in visual agnosia patients with bilateral ventral lesions,” *Cortex*, vol. 119, pp. 555–568, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010945219301017> 2, 3, 4
- [39] R. L. Klatzky and S. J. Lederman, “Multisensory texture perception,” *Multisensory object perception in the primate brain*, pp. 211–230, 2010. 2
- [40] M. Gori, L. Giuliana, G. Sandini, and D. Burr, “Visual size perception and haptic calibration during development,” *Developmental Science*, vol. 15, no. 6, pp. 854–862, 2012. [Online]. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-7687.2012.01183.x> 2, 3, 4
- [41] Y. Itaguchi, “Size perception bias and reach-to-grasp kinematics: An exploratory study on the virtual hand with a consumer immersive virtual-reality device,” *Frontiers in Virtual Reality*, vol. 2, p. 712378, 2021. 2
- [42] A. G. De Siqueira, R. Venkatakrishnan, R. Venkatakrishnan, A. Bhargava, K. Lucaites, H. Solini, M. Nasiri, A. Robb, C. Pagano, B. Ullmer, and S. V. Babu, “Empirically evaluating the effects of perceptual information channels on the size perception of tangibles in near-field virtual reality,” in *2021 IEEE Virtual Reality and 3D User Interfaces (VR)*. Lisboa, Portugal: IEEE, Mar. 2021, p. 1–10. [Online]. Available: <https://ieeexplore.ieee.org/document/9417639/> 2
- [43] F. Salehi, F. Pariafsai, and M. K. Dixit, “The impact of different levels of spatial cues on size perception: a spatial perception study of altered conditions,” in *International Conference on Human-Computer Interaction*. Springer, 2024, pp. 74–85. 2
- [44] J. K. Stefanucci, D. A. Lessard, M. N. Geuss, S. H. Creem-Regehr, and W. B. Thompson, “Evaluating the accuracy of size perception in real and virtual environments,” in *Proceedings of the ACM Symposium on Applied Perception*, 2012, pp. 79–82. 2
- [45] E. Lavoie, J. S. Hebert, and C. S. Chapman, “Comparing eye-hand coordination between controller-mediated virtual reality, and a real-world object interaction task,” *Journal of Vision*, vol. 24, no. 2, pp. 9–9, 2024. 2
- [46] Y.-C. Chen, A. Deza, and T. Konkle, “How big should this object be? perceptual influences on viewing-size preferences,” *Cognition*, vol. 225, p. 105114, 2022. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0010027722001020> 3
- [47] K. K. Ball, B. L. Beard, D. L. Roenker, R. L. Miller, and D. S. Griggs, “Age and visual search: Expanding the useful field of view,” *Journal of the Optical Society of America A*, vol. 5, no. 12, pp. 2210–2219, 1988. 3
- [48] K. Nakano, N. Isoyama, D. Monteiro, N. Sakata, K. Kiyokawa, and T. Narumi, “Head-mounted display with increased downward field of view improves presence and sense of self-location,” *IEEE Transactions on Visualization and Computer Graphics*, vol. 27, no. 11, pp. 4204–4214, 2021. 3
- [49] M. Kahrmanovic, W. M. Bergmann Tiest, and A. M. Kappers, “Discrimination thresholds for haptic perception of volume, surface area, and weight,” *Attention, perception, & psychophysics*, vol. 73, pp. 2649–2656, 2011. 4
- [50] M. Kahrmanovic, W. M. B. Tiest, and A. M. Kappers, “Haptic perception of volume and surface area of 3-d objects,” *Attention, Perception, & Psychophysics*, vol. 72, no. 2, pp. 517–527, 2010. 4
- [51] C. von Castell, H. Hecht, and D. Oberfeld, “Measuring perceived ceiling height in a visual comparison task,” *Quarterly Journal of Experimental Psychology*, vol. 70, no. 3, pp. 516–532, 2017. 4